

Production Scaling v7 Benchmark (300k calc/s)

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PAPER_930: Production Scaling v7 Benchmark (300k calc/s)

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Source: SCm phonon gap implementation (production_scaling_v7.py) **Calculator:** ProductionScalingV7BenchmarkCalc (CP4 #514) **CVW:** v2.0.0 compliant

Abstract

Production benchmark suite targeting 300,000 calculations/second, a 3x improvement over the v4 baseline (100k calc/s). Implements 8 benchmark kernels: 26-Layer Gravity summation, F_U_Bi_i Assembly, Phonon a_res computation, Jet M_jet(Gamma) evaluation, NS Spindown correction, GW190425 strain calculation, Full Pipeline v7 (all-in-one), and Vectorized Phonon Batch processing. Each kernel is timed independently and the aggregate throughput is compared against the 300k target. The benchmark validates that the new phonon physics modules (Session 211) maintain production-grade performance.

1. Core Equations

Section A: Benchmark Kernels

$$K1 : g = \sum_{k=1}^{26} \exp(-[SSq] * k/26) * (1 + 0.001 * k)$$

$$K2 : F_U_Bi_i = F_{U,Bi} * S_26 * [SSq] * E_{net}$$

$$K3 : a_{res} = (F_{U,Bi}/F_U) * Phi * S_26$$

$$K4 : M_{jet} = 1 + A_{jet} * \exp[-(Gamma - Gamma_0)^2 / (2 * sigma^2)]$$

$$K5 : \Omega_{dot_phonon} = \Omega_{dot} * (1 + Phi * S_26 * [SSq]/N)$$

$$K6 : h_U QFF = h_G R * D_{total} * \exp([SSq] * t/26)$$

$$K7 : Fullpipeline(K1 - K6_{sequential})$$

$$K8 : Vectorizedphononbatch(1000frequencies)$$

Section B: Performance Metrics

$$Rate = N_{iterations}/elapsed_{time}(calc/s)$$

Target : 300,000calc/saggregate

v4baseline : 100,000calc/s

Improvement factor : 3.0x

Section SM: SM Anchors

Python 3.14 runtime

Pure Python (no numpy requirement for core kernels)

Optional numpy vectorization for K8

2. Parameters

Parameter	Default	Description
n_iterations	10000	Benchmark iterations
target_calc_per_s	300000	Target throughput

3. Key Results

Kernel	Typical Rate (calc/s)	Status
26-Layer Gravity	~500k	PASS
Phonon a_res	~800k	PASS
Jet M_jet	~1M+	PASS
Full Pipeline	~200k	MARGINAL
Vectorized Batch	~400k	PASS

4. Physical Interpretation

The production scaling benchmark ensures that adding phonon physics modules does not degrade computational throughput below operational requirements. The 300k target represents the minimum rate needed for real-time simulation of 26-layer gravity across multiple astrophysical systems simultaneously (e.g., source2.cpp GUI parallel queries). Individual kernels exceed the target, but the full pipeline (all kernels sequential) approaches the limit, indicating that optimization of the pipeline serialization is the primary bottleneck.

5. References

- production_scaling_v4.py: Previous 100k baseline
- production_scaling_v7.py: Current 300k target implementation
- PAPER_922: M87 jet power curve (performance-critical MC sampling)

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25 \text{ THz}$ (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{vac} \sim 10^{-29} \text{ g/cm}^3$	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms → SCm vacuum → phonon ω_{SCm} → lattice resonance → F_{U,Bi_i} unified force → observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$$

VDS sub-ratio: 0.1

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (sub-threshold)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: 10^4 yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed